

Cases Cheat Sheet

Use Cases, Analysis, Verification

1. Use Case Definition

Functional goals.

```
package Cases_1UseCaseDefinition {
  private import ScalarValues::*;
  private import SysML::*;
  attribute def Vehicle;
  attribute def Person;
  /* Wrapped Snippet (Structure Context */)
  use case def DriveCar {
    subject vehicle : Vehicle ;
    actor driver : Person ;
    objective {
      doc /* Transport safely */
    }
  }
}
```

2. Test Case (Verification)

Verifying requirements.

```
package Cases_2TestCaseVerification {
  private import ScalarValues::*;
  private import SysML::*;
  attribute def Vehicle;
  attribute def Person;
  attribute def VerdictKind;
  /* Wrapped Snippet (Structure Context */)
  requirement brakeReq;
  requirement stoppingDistance;
  verification def TestBrakes {
    objective {
      verify brakeReq ;
      verify stoppingDistance ;
    }
    return verdict : VerdictKind ;
  }
}
```

3. Analysis Case

Evaluating properties.

```
package Cases_3AnalysisCase {
  private import ScalarValues::*;
  private import SysML::*;
  attribute def Vehicle;
  /* Wrapped Snippet (Structure Context */)
  analysis def FuelEconomy {
    subject vehicle : Vehicle ;
    objective {
      doc /* Estimate MPG */
    }
    return mpg : Real ;
  }
}
```

4. Case Usage

Instantiating a case.

```
package Cases_4CaseUsage {
  private import ScalarValues::*;
  private import SysML::*;
  attribute def Vehicle; attribute def Person;
  use case def DriveCar { subject vehicle : Vehicle; actor driver : Person; }
  part me : Person;
  /* Wrapped Snippet (Structure Context */)
  use case driveToWork : DriveCar {
    actor driver = me ;
  }
}
```